

DesignHive Admin

Chat Interface — Multi-Agent Test Cases

Generated May 20, 2026 · 10 Test Cases · Chat Agent (#9)

Each test case below is executed by typing the **User Message** directly into the Chat Interface (AI Assistant drawer). The Chat Agent (#9) resolves the request through its agentic loop, calling multiple backend agents in sequence. All 10 cases are designed to exercise cross-agent coordination.

TC-01

Generate + Schedule + Segment

USER MESSAGE

"Write a friendly re-engagement email about our new design toolkit and schedule it for all users inactive over 30 days, send it tomorrow at 9 AM."

AGENTS INVOLVED

Content Generation (2)	Segmentation (1)	Campaign Scheduler (3)
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EXPECTED AGENT FLOW

generate_content → segment_users (inactive, threshold=30) → schedule_campaign (tomorrow 09:00)

TC-02

Segment + Send Now + Report

USER MESSAGE

"Send the welcome email to all new users who signed up this week and give me a report once it's done."

AGENTS INVOLVED

Segmentation (1)	Campaign Scheduler (3)	Campaign Reporter (8)
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EXPECTED AGENT FLOW

segment_users (new_users) → send_campaign_now → get_campaign_report

TC-03

Stats + Conditional Segment + Send

USER MESSAGE

"How many users have never received an email? If it's more than 10, send them the getting-started template right now."

AGENTS INVOLVED

Segmentation (1)

Campaign Reporter (8)

Campaign Scheduler (3)

EXPECTED AGENT FLOW

segment_users (never_emaild) → get_email_stats (count check) → conditional send_campaign_now

Note: Tests conditional branching — agent must reason about a count before acting.

TC-04

Failure Recovery + Stats

USER MESSAGE

"Retry all the failed emails from the last 24 hours and then show me the updated delivery stats."

AGENTS INVOLVED

Failure Recovery (7)

Campaign Reporter (8)

EXPECTED AGENT FLOW

run_failure_recovery → get_email_stats (since_days=1)

TC-05

Re-engagement + Drip Enrollment

USER MESSAGE

"Find all users who haven't been active in 45 days and enroll them in the re-engagement drip sequence."

AGENTS INVOLVED

Segmentation (1)

Re-engagement (6)

Drip Sequence (4)

EXPECTED AGENT FLOW

run_reengagement (threshold=45) → segment_users (inactive, 45 days) → drip enrollment per matched user

TC-06

Generate + Custom Date Segment + Schedule

USER MESSAGE

"Create a professional email announcing our May sale, target users who signed up between March 1 and April 30, and schedule it for this Friday at 2 PM."

AGENTS INVOLVED

Content Generation (2)

Segmentation (1)

Campaign Scheduler (3)

EXPECTED AGENT FLOW

generate_content (professional tone) → segment_users (custom_date_range, Mar 1 – Apr 30) → schedule_campaign (Friday 14:00)

TC-07

List Scheduled + Cancel + Reschedule

USER MESSAGE

"Show me all upcoming scheduled campaigns. Cancel the Friday one and reschedule it for Monday at 10 AM with the same template and audience."

AGENTS INVOLVED

Campaign Scheduler (3)

EXPECTED AGENT FLOW

list_scheduled_campaigns → cancel (DELETE Friday job) → schedule_campaign (same template/segment, Monday 10:00)

Note: Tests state mutation — cancel then recreate with carried-over parameters.

TC-08

Stats → Generate Replacement + Send All

USER MESSAGE

"Which email template has the lowest success rate? Write a better version of it with an urgent tone and send it to everyone right now."

AGENTS INVOLVED

Campaign Reporter (8)

Content Generation (2)

Segmentation (1)

Campaign Scheduler (1)

EXPECTED AGENT FLOW

get_campaign_report (find worst template) → generate_content (urgent tone) → segment_users (all) → send_campaign_now

Note: Tests chained decisions — agent must identify the worst performer then act on it.

TC-09

Full Pipeline: Generate + Segment + Schedule + Report

USER MESSAGE

"Write a friendly email about our new asset library, send it to all users who've never been emailed, schedule it for today at noon, and report back with the results."

AGENTS INVOLVED

Content Generation (2)

Segmentation (1)

Campaign Scheduler (3)

Campaign Re

EXPECTED AGENT FLOW

generate_content → segment_users (never_emailed) → schedule_campaign (today 12:00) →
get_campaign_report

TC-10

Re-engagement + Failure Recovery + Stats (One Shot)

USER MESSAGE

"Run the re-engagement agent for inactive users, retry any emails that failed today, and then give me a full summary of today's email performance."

AGENTS INVOLVED

Segmentation (1)

Re-engagement (6)

Failure Recovery (7)

Campaign Re

EXPECTED AGENT FLOW

run_reengagement (triggers segmentation + send) → run_failure_recovery → get_email_stats
(since_days=1)

Note: Deepest chain — 4 distinct agents fire in sequence from a single message.

Agent Reference

#	Agent	File	Used in
1	Smart Segmentation	agents/segmentation.py	TC-01 02 03 05 06 08 09 10
2	Content Generation	agents/content_gen.py	TC-01 06 08 09
3	Campaign Scheduler	agents/scheduler.py	TC-01 02 03 06 07 08 09
4	Drip Sequence	agents/drip.py	TC-05

6	Re-engagement	agents/reengagement.py	TC-05 10
7	Failure Recovery	agents/failure_recovery.py	TC-04 10
8	Campaign Reporter	agents/reporter.py	TC-02 03 04 08 09 10
9	Chat Interface	agents/chat.py	All (entry point)